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Sr. No.	Practical List	Date	Sign
1.	**Data Pre-processing and Exploration**		
	a. Load a CSV dataset. Handle missing values, inconsistent formatting, and outliers.		
	b. Load a dataset, calculate descriptive summary statistics, create visualizations using different graphs, and identify potential features and target variables. Note: Explore Univariate and Bivariate graphs (Matplotlib) and Seaborn for visualization.		
	c. Create or explore datasets to use all pre-processing routines like label encoding, scaling, and binarization.		
2.	**Testing Hypothesis**		
	a. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a CSV file and generate the final specific hypothesis. (Create your dataset)		
3.	**Linear Models**		
	a. Simple Linear Regression — Fit a linear regression model on a dataset. Interpret coefficients, make predictions, and evaluate performance using metrics like R-squared and MSE.		
	b. Multiple Linear Regression — Extend linear regression to multiple features. Handle feature selection and potential multicollinearity.		
	c. Regularized Linear Models (Ridge, Lasso, ElasticNet) — Implement regression variants like LASSO and Ridge on any generated dataset.		
4.	**Discriminative Models**		
	a. Logistic Regression — Perform binary classification using logistic regression. Calculate accuracy, precision, recall, and understand the ROC curve.		
	b. Implement and demonstrate k-nearest Neighbor algorithm. Read the training data from a CSV file and build the model to classify a test sample. Print both correct and wrong predictions.		
	c. Build a decision tree classifier or regressor. Control hyperparameters like tree depth to avoid overfitting. Visualize the tree.		
	d. Implement a Support Vector Machine for any relevant dataset.		
	e. Train a random forest ensemble. Experiment with the number of trees and feature sampling. Compare performance to a single decision tree.		
5.	**Generative Models**		

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Sr. No.	Practical List	Date	Sign
	a. Implement and demonstrate the working of a Naive Bayesian classifier using a sample dataset. Build the model to classify a test sample.		
	b. Implement Hidden Markov Models using hmmlearn.		
6.	**Probabilistic Models**		
	a. Implement Bayesian Linear Regression to explore prior and posterior distribution.		
	b. Implement Gaussian Mixture Models for density estimation and unsupervised clustering.		
7.	**Model Evaluation and Hyperparameter Tuning**		
	a. Implement cross-validation techniques (k-fold, stratified, etc.) for robust model evaluation.		
	b. Systematically explore combinations of hyperparameters to optimize model performance (use grid and randomized search).		
8.	**Bayesian Learning**		
	a. Implement Bayesian Learning using inferences.		
9.	**Deep Generative Models**		
	a. Set up a generator network to produce samples and a discriminator network to distinguish between real and generated data. (Use a simple small dataset).		
10.	**Model Deployment**		
	a. Develop an API to deploy your model and perform predictions.		